CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (E & C) Examination

May 2013

EC – 309 Digital Communication

Date: 01-05-13, Wednesday **Time: 10:00 a.m. to 01:00 p.m.** **Maximum Marks: 70**

***Instructions*:**

(1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4)Rough work is done in the last page of main answer book; don't write anything on the question paper.

SECTION-I

Q-1 **A** Consider the analog signal $x(t) = 3\cos 100\pi t$ **3**

i) Determine the minimum sampling rate required to avoid aliasing.

ii) Suppose that the signal is sampled at the rate of Fs = 200 Hz, what is the discrete time signal obtained after sampling?

iii) Suppose that the signal is sampled at the rate of Fs = 75 Hz, what is the discrete time signal obtained after sampling?

iv) What is the frequency of a sinusoid that yields samples identical to those obtained in part (iii)?

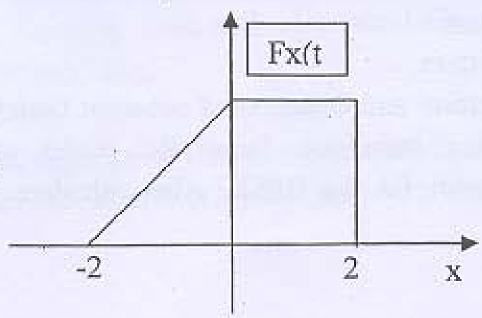
B Signal x(t) has a bandwidth of 12 kHz and its amplitude at any time is a random variable whose PDF is shown in the Figure below. We want to transmit this signal using a uniform PCM system. **4**

(i) Show that a=1/3.

(ii) Determine the power in x(t).

(iii) What is SQNR in decibels if a PCM system with 32 levels is employed?

(iv) What is the minimum required transmission bandwidth in Part 3?



C Explain the delta modulation stating its advantages and disadvantages. Also suggest the method to overcome these disadvantages. **4**

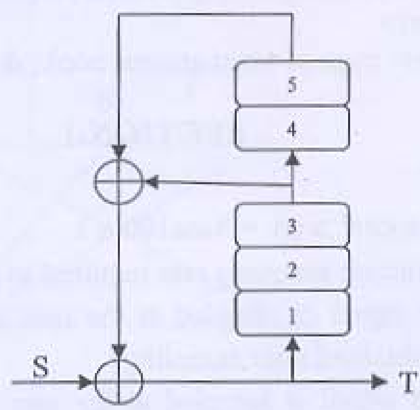
Q-2 Attempt any two.

A (i) Find the power spectrum density of polar and on-off where binary pulses are represented by full width rectangle pulse $p(t) = rect(t/T_b)$. **6**

(ii) Sketch roughly their PSD and finds their bandwidth.

B What is the Nyquist Criterion for zero ISI? Explain. **6**

C What is the need of scrambling? The data stream 101010100000111 is fed to scrambler shown in the figure below. Find the scrambler output assuming the initial content of the registers to be zero. **6**



Q-3 Attempt any two.

A Explain the generation and detection of coherent binary Frequency shift keying (BFSK) modulation technique. Draw the power spectrum density and a constellation diagram for the BFSK. Also calculate its minimum bandwidth requirement. **6**

B In a digital communication system, the bit rate of the NRZ data stream is 1 MbPS and the carrier frequency is 100 MHz. Find the symbol rate of transmission and bandwidth requirement of the channel in the following cases of different techniques used. **6**

(i) BPSK system

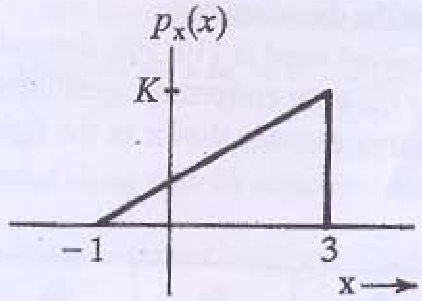
(ii) QPSK system

(iii)16-PSK system

C Explain the generation and detection of coherent binary Frequency shift keying (BPSK) modulation technique. Draw the power spectrum density and a constellation diagram for the BPSK. Also calculate its minimum bandwidth requirement. **6**

SECTION – II

Q-4 **A** Find the mean, mean square and the variance of the RV x whose PDF is given below: 3



B A random variable X has the probability density function 3

$$f_x(x) = \begin{cases} \frac{\pi}{16}\cos\frac{\pi x}{8} & -4 \le x \le 4 \\ 0 & \text{otherwise} \end{cases}$$

Find its mean value,second moment and its variance.

C A dice is tossed once. Let X denotes its outcome. 3

(1) Find the cumulative distribution function and draw the graph of the CDF

(2) Find the probability $P(3 \le X < 5)$

D For certain binary asymmetric channel, it is given that $P_{y/x}(0|1) = 0.1$ and $P_{y/x}(1|0) = 0.2$,Where x is transmitted digit and y is received digit. If $P_x(0) = 0.4$, determine $P_y(0)$ and $P_y(1)$. 2

Q-5 **Attempt any two.**

A A Zero memory source emits six messages with probabilities 0.3,0.25,0.15,0.12,0.1and 0.8. Find the optimum Huffman code and quaternary Huffman code. Compare both codes by calculating the average word length ,the efficiency and redundancy. 6

B Find the channel capacity of Binary Symmetric Channel (BSC). 6

C For (6,3) systematic linear block code, the parity check digits c_4,c_5 and c_6 are 6

$c_4=d_1+d_2+d_3$

$c_5=d_1+d_2$

$c_6=d1+d_3$

(1) Construct the appropriate generator matrix

(2) Construct the code generated by this matrix

(3) Determine error correcting and detecting probabilities for the code

Q-6 **Attempt any two.**

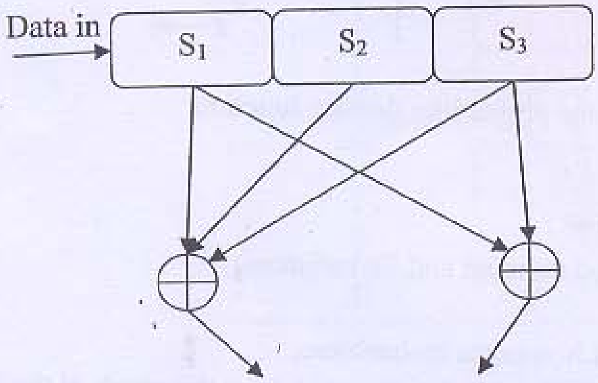
A Construct a systemetic (7,4) cyclic code using generator polynomial $g(x) = x^3+x+1$. **6**

(1) Construct the decoding table .

(2) If the received word is 1101100, determine the transmitted data word.

(3) What are the error correcting capabilities of this code.

B For the convolution encoder shown in the figure below, draw its state diagram, Tree diagram. **6**



C (1) Show that the channel capacity of being an ideal AWGN Channel with infinite bandwidth B is given by $C_\infty = 1.44\frac{S}{\eta}$, Where noise power $N = \eta B$, S is the average signal power and $\eta/2$ is the power spectrum density of white Gaussian noise. **3**

(2) three error correcting (23,12) Golay code is a cyclic code with a generator polynomial $G(x)=x^{11}+x^9+x^7+x^6+x^5+x+1$.Determine the code words for the data vectors 000011110000,101010101010 **3**